Per-unit and synchronous reactance

Cheat sheet · Day 13 · Thu 5 Nov · pack M10

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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PER-UNIT (Ch8.3)        pu = actual / base      choice of base never changes real ans
 Primary:   S_B = rating (total 3ph VA),  V_B = rated LINE voltage
 Derived:   I_B = S_B / (sqrt3 V_B)   [1ph: S_B/V_B]
            Z_B = R_B = X_B = V_B^2 / S_B      P_B = Q_B = S_B
 In pu:     line V pu = phase V pu; total S pu = phase S pu; NO sqrt3 or 3
            S = V I*,  P = V I cos(phi)
 Back:      V_line = V(pu) x V_B ;  I(A) = I(pu) x I_B ;  Z(ohm) = Z(pu) x Z_B

SYNC MACHINE MODEL (Ch8.4)   I out of machine (generator convention)
 E0 = V + I (R + jXs)          V = V<0 reference ; delta = angle of E0
 Motor (I_m into machine): E0 = V - I_m (R + jXs) ; delta < 0
 Field current IF sets E0 (and Q); rotor angle sets P  (see M11)

Xs FROM TESTS (Ch8.5) - SAME IF for both readings
 |Zs| = (V_OC,line / sqrt3) / I_SC        Xs = sqrt(|Zs|^2 - R^2)
 Xs(pu) = Xs / Z_B           slopes: use IF = 1 A
 IF for E0 = 1 pu (synchronisation) = IF_test x V_B / V_OC
 IF = E0(pu) x IF(E0 = 1 pu)    (unsaturated)

V and I FROM LOAD
 V = V<0 pu (1.05 = 5% above, 0.97 = 3% below)
 S = P/pf  or sqrt(P^2+Q^2)  (pu on machine rating)
 I = (S/V) < -acos(pf)  lag / delivering Q
 I = (S/V) < +acos(pf)  lead / absorbing Q      or  I = (S/V)*
 Check: MW is not MVA; use actual V in S/V; answer in ohms or pu as asked
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